Dynasonics®
Ultrasonic Flow Meters

Transducers

Easy Rail Mounting System (JZ/KZ)

INTRODUCTION

This document explains how to install transit time ultrasonic transducers with Easy Rail mounting systems.

The Easy Rail mounting system can be installed on vertical or horizontal pipes. For horizontal applications, install the system on the side of the pipe.

⚠ WARNING

EQUIPMENT SHALL BE PROTECTED FROM UV LIGHT.

⚠ WARNING

EQUIPMENT TO BE PROTECTED FROM IMPACT.

⚠ WARNING

THIS EQUIPMENT INCLUDES EXTERNAL NON-METALLIC PARTS. THE USER SHALL THEREFORE ENSURE THAT THE EQUIPMENT IS NOT INSTALLED IN A LOCATION WHERE IT MAY BE SUBJECTED TO EXTERNAL CONDITIONS (SUCH AS HIGH-PRESSURE STEAM) WHICH MIGHT CAUSE A BUILD-UP OF ELECTROSTATIC CHARGES ON NON-CONDUCTING SURFACES. ADDITIONALLY, CLEANING OF THE EQUIPMENT SHOULD BE DONE ONLY WITH A DAMP CLOTH.

⚠ WARNING

DO NOT CONNECT OR DISCONNECT WHEN ENERGIZED.

SPECIFICATIONS

- Voltage: 30 V
- Current: 0.375 A
- Power: 3.15 W

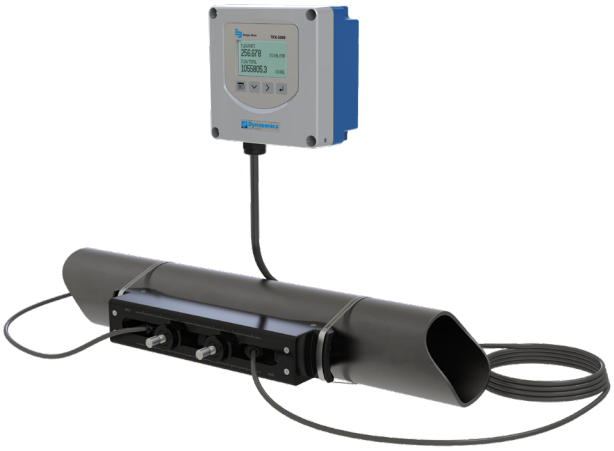
PRE-INSTALLATION REQUIREMENTS

NOTE: Protect all parts until installation is complete.

Program the Meter

Before installing the Easy Rail mounting system, you must select the optimum transmission mode, enter the site information and enter the fluid and pipe properties into the ultrasonic flow meter. For detailed instructions, refer to the user manual for your flow meter.

- Dynasonics TFX-500w, *TTM-UM-02537-EN*
- Dynasonics TFX-5000, *TTM-UM-02222-EN*
- Dynasonics DXN-5P, *HYB-UM-04661-EN*



Select a Pipe Location for the Easy Rail Mounting System

Select a location for the Easy Rail mounting system on a section of pipe that has at least 10 pipe diameters upstream of the transducers and 5 pipe diameters downstream. See *Figure 2 on page 2*.

For example, if a 3 inch pipe is being measured, the minimum upstream pipe in front of the mounting rail should be 30 in. and the minimum downstream pipe behind the transducers should be at least 15 in.

Pipe runs shorter than the minimums may sometimes be used with reduced accuracy. There is no way to determine how much accuracy is sacrificed without doing in-field testing. For installations where the 10/5 pipe diameters rule cannot be followed, divide the total length of available straight pipe into thirds and mount the rail with 2/3 of the pipe upstream and 1/3 of the pipe downstream.

A full pipe is absolutely essential for making accurate flow measurements. The flow meter cannot determine if the pipe is full or not. If the pipe is partially full, the meter will over-report the amount of flow by the percentage of the pipe that is not filled with liquid or may not detect any flow.

Install the mounting system in an area where the transducers will not be inadvertently bumped or disturbed.

Avoid installations on downward flowing pipes unless adequate downstream head pressure is present to overcome partial filling of—or cavitation in—the pipe.

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Installation Guide

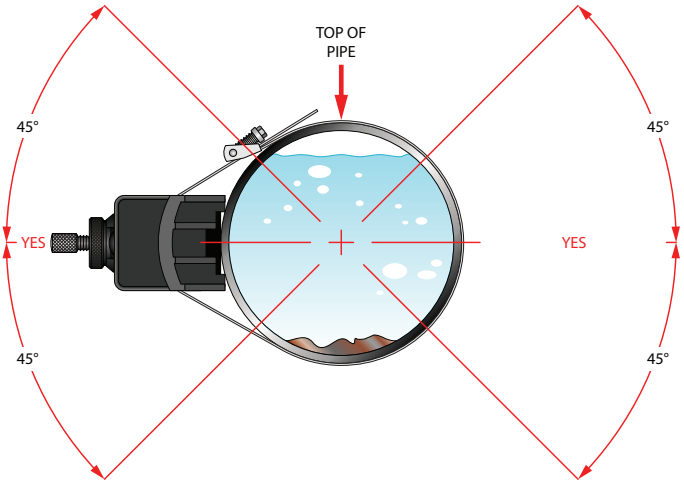


Figure 1: Transducer positioning

Piping Configurations and Transducer Positioning

Figure 2 shows the number of pipe diameters required downstream and upstream of the transducers for various piping configurations.

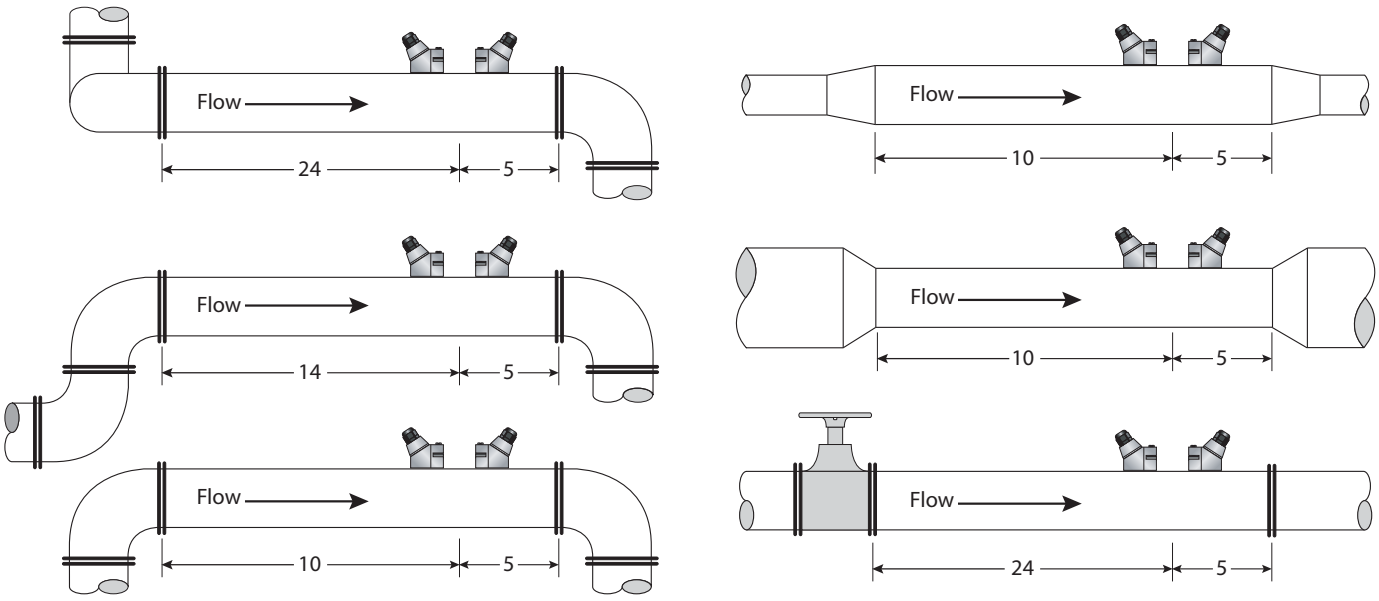


Figure 2: Piping configuration and transducer positioning

The system will provide repeatable measurements on piping systems that do *not* meet these pipe diameter requirements, but the accuracy of the readings may be influenced.

Partially-Filled Pipe Situations

In some locations, the process pipe may be momentarily only partially filled. Examples include: lack of back pressure, insufficient line pressure and gravity flow applications.

To eliminate these situations:

- Do not install the transducers at the highest point of the pipeline.
- Do not install the transducers in a vertical, downward flow section of pipe.
- Always position the ON/OFF valves on the downstream side of the transducers.

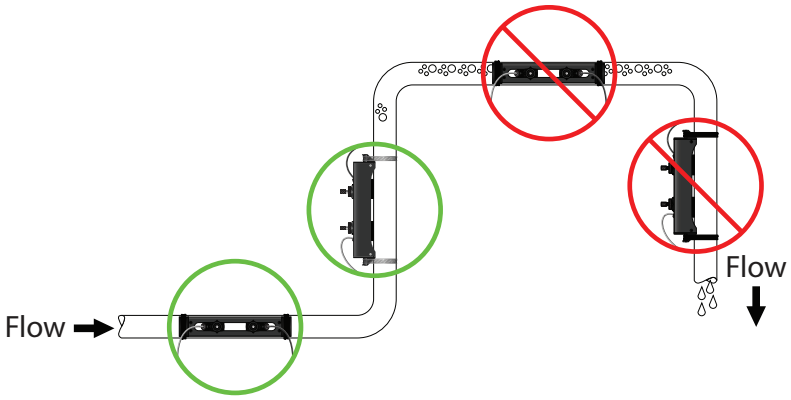
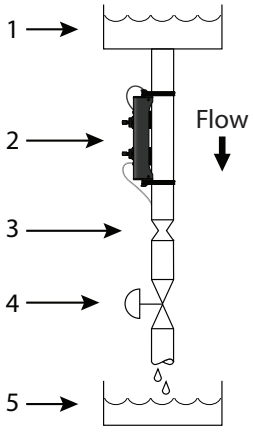


Figure 3: Transducer orientation



1	Supply Tank
2	Transducers
3	Pipe Restriction
4	Valve
5	Batching Tank

Figure 4: Transducer vertical mount, flow down

Pipe Material

Pipes must be good conductors of ultrasound and not block or scatter the signal. Most pipe materials will allow the signal to pass through. They include: stainless steel, PVC, CPVC, ABS, polypropylene, PVDF, copper, copper nickel, ductile iron and aluminum.

Wound fiberglass and concrete pipes typically trap air and are not suitable for these transducers. Some galvanized pipes may scatter the ultrasonic signal. Sometimes, relocating the transducers or mounting the transducers in a Z mode (1 transverse) will improve the signal strength.

Any liners in the pipe need to adhere to the pipe walls. Total wall thickness and liner should not exceed 10 in. (254 mm). New mortar-lined ductile iron pipes may have air trapped in the lining initially. Letting the pipe soak will allow water to displace the air and allow the signal to pass through.

Paint with good adhesion to the pipe typically allows the ultrasonic signal to pass through. If there is blistering or peeling paint, sand the paint off before installing the transducers.

NOTE: Do not mount the transducers over a welded pipe seam.

Test the signal strength before permanently installing the transducers with RTV or sealant.

INSTALLATION PROCEDURE

In the *Setup* > *Meter* or *Quick Start* menu, enter the mounting path, fluid and pipe properties into the transmitter or handheld. All moving parts are already mounted on the rail. During installation, make sure the rail is firmly positioned on the pipe and all thumbscrews are hand-tight. See *Figure 5*.

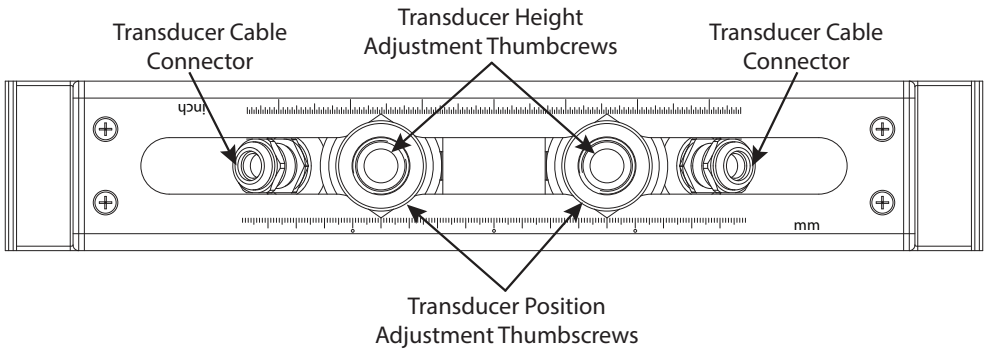


Figure 5: Easy rail top view

1. Clean the surface of the pipe. If the pipe has external corrosion or dirt, wire brush, sand or grind the mounting location until it is smooth and clean. Paint and other coatings, if not flaked or bubbled, need not be removed. Plastic pipes typically do not require surface preparation other than soap and water cleaning.
2. Loosen the position adjustment thumbscrews and slide the transducers along the scale to set the distance between the them. The top of the rail shows the distance in millimeters and inches.
3. Tighten the position adjustment thumbscrews hand-tight to lock the transducers in place.
4. Place couplant between the transducer and pipe. If using a couplant pad, it is not necessary to use a couplant paste or gel. Check that the pad covers the entire contact area between the transducer and pipe.
   If using a couplant paste, place approximately 1/2 inch (12 mm) thick, on the flat face of the transducers. See *Figure 6*. Generally, a silicone-based grease is used as an acoustic couplant, but any good quality grease-like substance that is rated to not flow or shrink at the operating temperature of the pipe is typically acceptable.
   To check the condition of the couplant, monitor for any decreases in signal strength and check for any physical changes to the couplant.

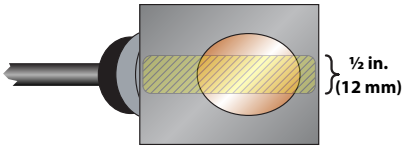


Figure 6: Transducer bottom view

Conditions	Couplant	Timeframe [1]
Pipe surface temperature under 350 °F (176 °C), dry or submerged, permanent mount	P.N. 70823-001 couplant pad for large pipes P.N. 70823-002 couplant pad for medium pipes	Product life
Pipe surface temperature under 130 °F (55 °C), dry pipe	P.N. D002-2011-001 or D002-2011-008 Dow Corning 111 Grease	1 year
Pipe surface temperature under 350 °F (177 °C), dry or submerged, permanent mount	P.N. D002-2011-002 Dow Corning 732 multi-purpose sealant	Product life
Pipe surface temperature under 350 °F (177 °C), dry pipe	P.N. D002-2011-011 or D002-2011-012 Dow Corning 340 heat sink compound	Product life
Pipe surface temperature under 350 °F (177 °C), dry pipe, silicone not permitted	P.N. D002-2011-009 Molykote G-N; non-silicone	1 year
Pipe surface temperature under 120 °F (49 °C), dry pipe	P.N. D002-2011-014 Aquasonic 100 water soluble ultrasound transmission gel	Less than 4 hours

[1] Timeframes are based on conditions where the transducers and couplant are not disturbed.
To check the condition of the couplant, monitor for any decreases in the signal strength and check for any physical changes to the couplant.

5. Place the mounting rail on the pipe and fasten it with metal banding straps. See *Figure 7*.
   a. Slide the strap into the arched groove on the end of the transducer.
   b. Wrap the strap around the pipe.
   c. Slide the free end of the strap into the end clip of the strap with the screw at 90 degrees to the strap.
   d. Pull the strap through until it loosely fits around the pipe.
   e. Rotate the screw so it is parallel to the strap and tighten the screw slightly to help hold the transducer onto the pipe.

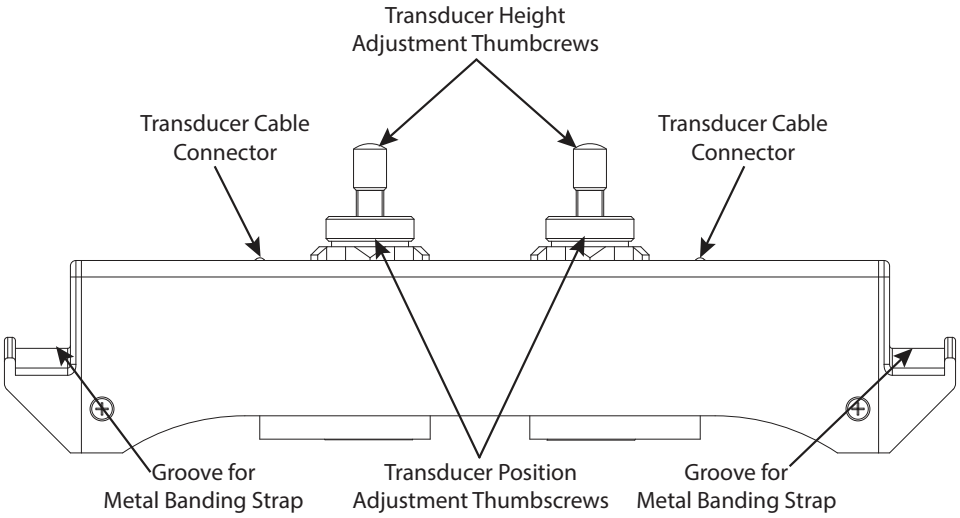


Figure 7: Easy rail side view

6. Turn the transducer height adjustment thumbscrews clockwise to lower the transducers onto the pipe surface.
7. Verify that the transducer is true to the pipe and all air is expelled out of the gap between the transducer faces and the pipe as necessary.
8. Route the transducer cables back to the flow meter location, avoiding high voltage cable trays and conduits.

Wiring/Cable Connections

After installing the transducers, connect the transducers to the transmitter or handheld with the cables provided. Refer to manuals for instructions. For DXN-5P portable handheld, connect the transducers to the 3-pin TRANSIT TIME connectors on the handheld using the yellow cables.

- Dynasonics TFX-500w, *TTM-UM-02537-EN*
- Dynasonics TFX-5000, *TTM-UM-02222-EN*
- Dynasonics DXN-5P, *HYB-UM-04661-EN*

Wiring the Transducer to TFX-500w or TFX-5000 Transmitter

NOTE: Submersible transducer cables are larger diameter. Each cable requires a separate conduit hole. The standard yellow cable and high temperature cables are small enough to use a single cable gland with a 2-hole grommet.

NOTE: Transducer cables have two wire-color combinations. For the blue and white combination, the blue wire is positive (+) and the white wire is negative (–). For the red and black combination, the red wire is positive (+) and the black wire is negative (–). The transducer wires are labeled to indicate which pair is upstream or downstream.

NOTE: For transducers with only two wires per cable, omit the ground connection on pins 3 and 6.

1. Guide the transducer terminations through a conduit hole in the bottom of the enclosure.
2. Secure the transducer cable using the supplied conduit nut (if flexible conduit was ordered with the transducer).
3. Install the ferrite to the cable:
   a. To open the ferrite, pull the fastener away from the body of the ferrite.
   b. Wrap the cable tightly around half of the ferrite and place the cable into the groove.
   c. Snap the ferrite shut.



4. The terminals within the transmitter are screw-down barrier terminals. Connect the wires at the corresponding screw terminals in the transmitter. Observe upstream and downstream orientation and wire polarity. For the TFX-500w transducer, refer to *Figure 8* and for the TFX-5000 transducer, refer to *Figure 9*.

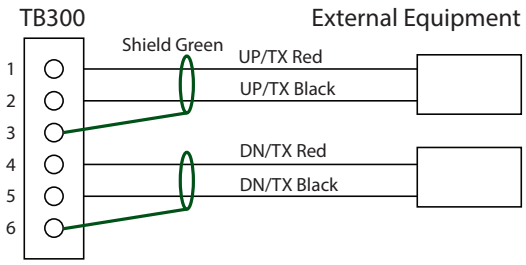


Figure 8: TFX-500w

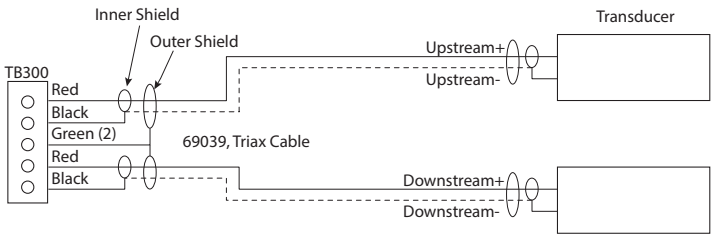


Figure 9: TFX-5000

Signal Strength

At the transmitter or handheld, check that the signal strength is good. If the signal strength is low, check the couplant between the transducers and pipe, and verify the pipe is full. If after verifying the pipe is full and the signal strength is still low, relocate the transducers and retest.

Transducer Spacing

Spacing for V-Mount Medium Pipe Transducers on ASME/ANSI Pipes with Potable Water

After the mounting path, fluid and pipe properties are entered into the transmitter, the transmitter calculates the proper transducer spacing and displays the spacing in Setup > Meter > Spacing menu or the Quick Start menu. *Table 1* lists the transducer spacing for common ASME/ANSI pipe sizes, materials and schedules.

NOTE: JZ transducers accommodate only up to 6 in. pipes.

Pipe Size	Pipe Material	Schedule	Spacing
3 in.	CPVC or PVC	Schedule 40, Standard	2.83 in.
		Schedule 80	2.81 in.
	Carbon Steel	Schedule 40, Standard	3.07 in.
		Schedule 80	3.15 in.
	Stainless 316/316L	Schedule 40, Standard	3.10 in.
		Schedule 80	3.18 in.
	Stainless 304	Schedule 40, Standard	3.03 in.
		Schedule 80	3.10 in.
	Copper	Type K	2.53 in.
4 in.	CPVC or PVC	Schedule 40, Standard	3.52 in.
		Schedule 80	3.51 in.
	Carbon Steel	Schedule 40, Standard	3.79 in.
		Schedule 80	3.89 in.
	Stainless 316/316L	Schedule 40, Standard	3.81 in.
		Schedule 80	3.92 in.
	Stainless 304	Schedule 40, Standard	3.74 in.
		Schedule 80	3.83 in.
	Copper	Type K	3.22 in.
6 in.	CPVC or PVC	Schedule 40, Standard	5.19 in.
		Schedule 80	5.17 in.
	Carbon Steel	Schedule 40, Standard	5.50 in.
		Schedule 80	5.66 in.
	Stainless 316/316L	Schedule 40, Standard	5.53 in.
		Schedule 80	5.70 in.
	Stainless 304	Schedule 40, Standard	5.45 in.
		Schedule 80	5.58 in.
8 in.	CPVC or PVC	Schedule 40, Standard	7.00 in.
		Schedule 80	6.97 in.
	Carbon Steel	Schedule 40, Standard	7.37 in.
		Schedule 80	7.54 in.
	Stainless 316/316L	Schedule 40, Standard	7.40 in.
		Schedule 80	7.58 in.
	Stainless 304	Schedule 40, Standard	7.31 in.
		Schedule 80	7.45 in.
10 in.	CPVC or PVC	Schedule 40, Standard	8.43 in.
		Schedule 80	8.40 in.
	Carbon Steel	Schedule 40, Standard	8.84 in.
		Schedule 80	9.07 in.
	Stainless 316/316L	Schedule 40, Standard	8.88 in.
		Schedule 80	9.13 in.
	Stainless 304	Schedule 40, Standard	8.78 in.
		Schedule 80	8.97 in.
12 in.	Carbon Steel	Schedule 40, Standard	10.23 in.
		Schedule 80	10.53 in.
	Stainless 316/316L	Schedule 40, Standard	10.27 in.
		Schedule 80	10.59 in.
	Stainless 304	Schedule 40, Standard	10.16 in.
		Schedule 80	10.40 in.

Table 1: Transducer spacing

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